Facial Fidelity using Machine Learning

Problem statement:

Current facial recognition systems struggle with blurry or low-light images, leading to unreliable identification in real-world applications. This results in critical failures in security and verification, as traditional methods demand high-quality visual inputs. This project addresses these limitations by developing a machine learning-based facial fidelity system to ensure accurate individual recognition from degraded images, significantly enhancing robustness.

Real World Scenario: In incidents like the Pulwama attack, authorities often face challenges in identifying attackers due to limited visual data captured from CCTV cameras. These images are typically blurred, shaky, or affected by poor lighting. The goal is to process such low-quality images and accurately match them with existing records in the shortest possible time to support timely and effective response.

Introduction:

Traditional facial recognition struggles with blurry or low-light images, impacting identification accuracy. This project introduces a machine learning-based facial fidelity system to reliably identify individuals from degraded visual inputs. Advanced image enhancement algorithms improve input quality for robust recognition. Deep learning models are optimized for identifying persons from challenging, real-world low-fidelity images. High accuracy in adverse conditions provides a reliable solution for critical identification tasks.

Objectives:

1. Input Layer : Degraded Image Capture Sources: CCTV footage, mobile cameras, surveillance systemsConditions: low-light, motion blur, poor resolution

2. Preprocessing & Enhancement Layer : Image PreprocessingNoise reduction (Gaussian/Median filtering)Histogram equalization (for low-light)Image Enhancement Model (Deep Learning)Super-Resolution CNN / GAN (e.g., ESRGAN) Deblurring Network (Restormer, DeblurGAN)Low-light enhancement (Zero-DCE, EnlightenGAN)→ Outputs high-fidelity face images

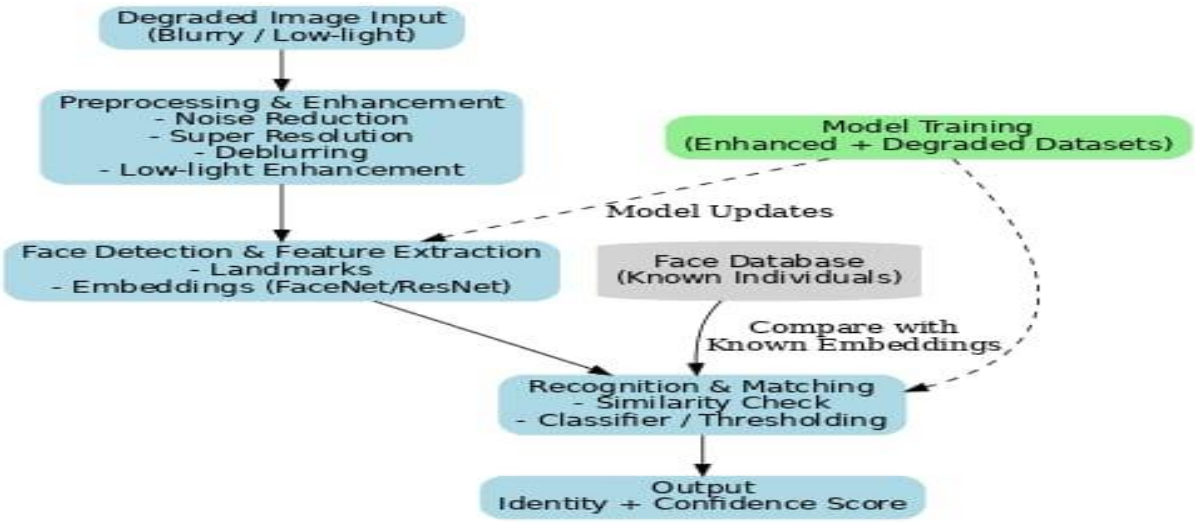
3. Feature Extraction Layer : Face Detection (MTCNN / RetinaFace / YOLO)Facial Landmark Extraction (eyes, nose, mouth keypoints)Embedding GenerationDeep CNN (ResNet, VGGFace, FaceNet)Converts face into numerical feature vector

4. Recognition & Matching Layer : Classifier / Metric LearningCosine similarity / Euclidean distance with known embeddingsOR ML classifier (SVM, Softmax, etc.)Identity MatchingCompare enhanced face embeddings against database of known individuals

5. Output Layer : Recognized Identity / Match Confidence ScoreUnidentified if below thresholdLogs & Alerts for critical systems (e.g., security use cases)

6. Database / Training Layer : Face Database (known individuals)Model TrainingDataset with degraded + enhanced imagesSupervised learning with augmentation (blur, noise, low-light)Continuous Learning (improve over time with new inputs

Methodology:



1. Input Acquisition Capture degraded face image (CCTV, low-light, blurred input).

2. Preprocessing & Enhancement Noise removal Deblurring Super-resolution / Low-light enhancement

3. Face Detection Locate face region in enhanced image.

4. Feature Extraction Extract facial landmarks & embeddings (deep learning models).

5. Recognition & Matching Compare extracted features with database embeddings. Calculate similarity/confidence score.

6. Decision & Output If similarity ≥ threshold → Recognized Identity + Confidence Score Else → Mark as Unknown

7. Database & Model Update (Feedback Loop) Store new samples (if authorized). Retrain model periodically with new degraded/enhanced pairs.

Expected Outcome:

A robust facial recognition system that works even with blurred, noisy, or low-light images. Enhanced image quality using advanced preprocessing & deep learning models. High accuracy in identification compared to traditional methods under degraded conditions. Reliable recognition

with confidence score for each prediction. A secure face database for storing known identities and embeddings. Scalable solution that can be integrated into surveillance, security, and authentication systems. Demonstration of improved performance metrics (accuracy, precision, recall, F1-score) on challenging datasets. Contribution towards a real-world problem solution: enabling critical identification in adverse environments.

Applications:

- **Night-time Surveillance Systems:** Enhances security camera performance in low-light environments like streets, parking lots, or warehouses.
- **Smart Home Security:** Accurately detects and recognizes faces even in dimly lit areas for doorbell cameras or indoor monitoring.
- **Border and Airport Security:** Improves identity verification during night operations or in poor lighting conditions.
- **Attendance Systems:** Enables accurate face-based attendance in educational institutions and workplaces with variable lighting.
- **Crime Investigation Support:** Helps law enforcement recognize suspects from low-quality or blurred CCTV footage.
- **Mobile Face Unlock Systems:** Improves performance of facial authentication on smartphones in dark or shaky conditions.

Conclusion

Robust face recognition achievable even in poor image conditions.

Preprocessing + embeddings significantly improve accuracy.

Method can be extended to real-world security, surveillance, and authentication systems.